

DAY 2: POWER SUPPLY SYSTEMS

Task 1 Real-world Device Using SMPS & UPS

1. Devices Using SMPS (Switched Mode Power Supply)

1. **Desktop Computer (PC Tower):** Converts high-voltage AC power into multiple regulated DC voltages (+3.3V, +5V, +12V) required by internal components like motherboard, CPU, GPU, and hard drives.
2. **LED / Smart Television:** Uses an internal SMPS unit to step down high AC mains voltage into lower DC voltages, producing very little heat and saving energy.
3. **Mobile Phone Charger & Laptop Adapter:** Uses compact SMPS switching technology to convert AC wall current into safe DC power (5V, 9V, 12V, or 20V) for battery charging.
4. **CCTV Power Supply Unit:** Provides continuous, stable 12V DC power to multiple security cameras simultaneously over long cables.

2. Devices Using UPS (Uninterruptible Power Supply)

1. **Wi-Fi Router / ONT Box:** Uses mini DC-UPS to ensure internet connectivity remains active without interruption during electricity power cuts.
2. **Desktop Workstation:** Connected to a UPS to provide instant battery backup during grid failure, allowing users to save work and safely shut down the system.
3. **Hospital & Medical Life-Support Systems:** Uses double-conversion Online UPS systems to deliver uninterrupted power with zero transfer delay during emergency situations.

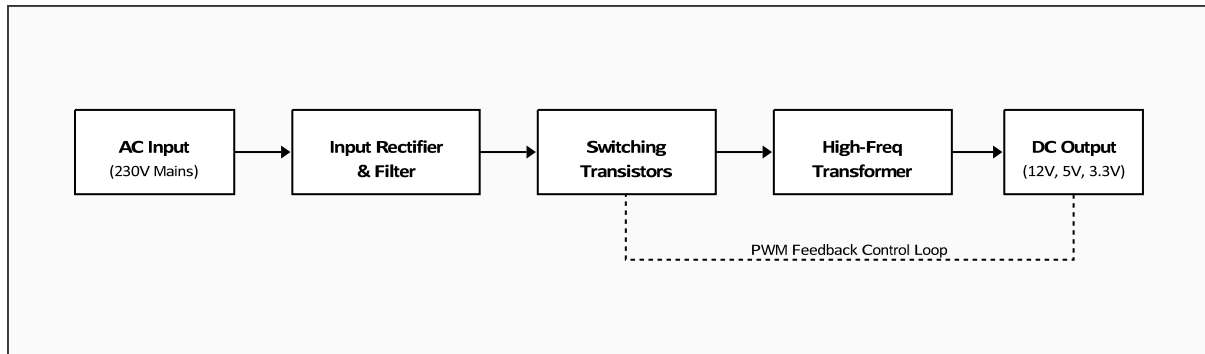
3. Comparison Table: SMPS vs. UPS

Feature	SMPS (Switched Mode Power Supply)	UPS (Uninterruptible Power Supply)
Primary Function	Converts high AC voltage to steady DC voltage.	Provides temporary power backup from internal battery.
Energy Storage	No energy storage capability (needs constant AC input).	Stores electrical energy inside rechargeable batteries.
Placement	Built inside devices or used as inline power adapters.	Connected between wall socket and equipment.
Main Advantage	High energy efficiency, lightweight, and compact design.	Instantaneous power backup and surge protection.

Task 2: Internal Architecture & Block Diagram Of SMPS

1. Block Diagram of Desktop SMPS

The diagram below shows the basic operational stages of an SMPS power supply:



2. Explanation of Main Internal Components

- **EMI/RFI Filter & Fuse:** Protects the power supply from electrical spikes and voltage surges, and prevents high-frequency interference from entering the power line.
- **Bridge Rectifier & Filter Capacitors:** Converts high-voltage AC mains power into DC voltage (approx. 300V DC) and smoothes out ripples.
- **Switching MOSFETs:** Switches high-voltage DC current on and off thousands of times per second (high frequency switching).
- **High-Frequency Transformer:** Steps down high-voltage AC to low-voltage AC efficiently using a ferrite core.
- **Output Rectifier Diodes & Filters:** Converts high-frequency low-voltage AC into clean DC outputs (+12V, +5V, +3.3V).
- **PWM Controller IC:** Controls output voltage accuracy by adjusting switching pulse widths according to load requirements.

Task 3: Power Connector & Voltage Specifications

1. Identification of Power Connectors

Connector Name	Powered Components	Handling & Safety Precautions
1. ATX 24-Pin Connector	Motherboard, RAM slots, PCIe slots.	Press plastic lock latch down before pulling. Pull straight upward without twisting wires.
2. EPS / CPU 4+4 Pin	Processor (CPU) power connector.	Verify orientation of split 4+4 connector before plugging into motherboard header.
3. PCIe 6+2 Pin Connector	Dedicated Graphics Cards (GPU).	Combine 6-pin and 2-pin clips tightly before inserting into GPU socket.
4. SATA Power Connector	SATA SSDs, Hard Drives, Optical Drives.	Has L-shaped plastic notch. Align gently to avoid breaking internal connector pin.
5. Legacy Molex 4-Pin	Case fans, older IDE hard drives, controllers.	Align internal metal pins carefully; do not apply strong force if resistance occurs.

2. Wire Color Code and Output Voltage Table

Wire Color	Voltage Level	Used For / Powered Components
Yellow	+12V DC	High power parts: CPU power VRM, Graphics Card, Fans, Hard Drive motors.
Red	+5V DC	Logic circuits, SATA SSDs, HDD logic boards, USB ports.
Orange	+3.3V DC	Motherboard chipsets, RAM modules, PCIe expansion logic.

Black	Ground / COM (0V)	Common reference return wire for all electrical circuits.
Green	PS_ON# (Power On Signal)	Triggers SMPS startup when connected to Ground wire.
Purple	+5VSB (Standby)	Provides constant +5V power when PC is turned off (for LAN boot, USB power).